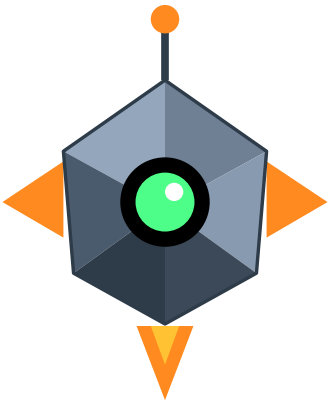
BOOK 1

A Useful Wrong Answer

Estimates that know how wrong they are





MULTIX // HELLO

I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

The Ruler That Ran Long



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. A circle is *ready* when everything needed for its calculation is known and its operation is allowed by the current rules. To **pop** a ready circle, replace it with its value and leave everything outside it alone. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Mara had a long board and a plan. She wanted a row of pegs for hanging coats, with each peg 7/8 of an inch from the next. The ruler in her toolbox was marked in whole inches, and 7/8 is awkward to find on a ruler like that. So she did the easy thing: she made a pencil mark at every whole inch. Mark 1 at 1 inch, mark 2 at 2 inches, mark 3 at 3 inches, and on down the board.

Then she looked at her plan again and groaned. Every mark was too far along. Her marks were 1 inch apart, and they were supposed to be 7/8 inch apart.

She could have erased everything and started over. Instead she asked a better question: how wrong is each mark, and can I fix them all without measuring again?

STEP 1 Find the mistake

Each step was supposed to be 7/8 inch, and she made it 1 inch. Since 1 is the same as 8/8, each step was too long by one eighth of an inch.

Now look at what that does as the marks pile up.

Mark	Where it landed	Where it should be	Move it back by
1st	1	7/8	1/8
2nd	2	1 3/4	1/4
3rd	3	2 5/8	3/8
4th	4	3 1/2	1/2

Look down the last column. It goes 1/8, 2/8, 3/8, 4/8. Every mark carries one more eighth of mistake than the mark before

it. So the fix for mark number n is simple: move it back by n eighths.

Optional, for brave readers: you can write the true position of mark n like this: start with n inches, then subtract n/8. That gives 7n/8.

Try it: The 8th mark needs eight eighths removed. Eight eighths make exactly one inch, so the 8th mark belongs at 7 inches, not 8. Check it with a real ruler.

IN CIRCLES

Mara can write the mistake with circles. Mark 4 landed at (4 × 1) inches, but it should be at (4 × 7/8) inches. To find how far back it must move, subtract one circle from the other, then *redraw* so that the 4 sits outside a single circle:

Trust check. There are three kinds of fixes in this chapter.

TRACE

```
( ( 4 × 1 ) − ( 4 × 7/8 ) )
redraw  →  ( 4 × ( 1 − 7/8 ) )
pop ( 1 − 7/8 )  →  1/8
( 4 × 1/8 )
pop  →  1/2
```

The redraw keeps the value, because four lots of 1 minus four lots of 7/8 is four lots of the difference. After the redraw, the whole mistake sits in one small circle, (1 − 7/8), which pops to 1/8. Every other mark just multiplies that 1/8 by its own number: (1 × 1/8), (2 × 1/8), (3 × 1/8), and so on. That is the last column of the table.

STEP 2 What if you only guessed?

Now change the story. Suppose Mara does not know the exact spacing. She thinks it is about 7/8, and she uses that guess to fix her marks.

Her uncle has made a sample board with the pegs correctly placed. Mara lays the sample beside her board, lines up the zero ends, and measures from that same end. Her guess says the 16th mark belongs at 14 inches (16 times 7/8). But the 16th peg on the sample board is at 13 3/4 inches. Her guess was a quarter inch too long.

If the sample board's pegs are evenly spaced, that quarter inch can be shared out over sixteen steps, about 1/64 of an inch each. One tiny error is invisible. Sixteen of them add up to something you can see.

So the guess was useful, because it got her very close, but it was still a little off, and the mistake grows as you go down the board. If the spacing really is even, the 32nd mark would be about a half inch off. The lesson: **if you are working from**

an estimate, check a mark far from the start against something you trust. That is where a small error shows up.

There is another way to see what went wrong with the guess. Mara's guess for mark 16 is a circle she popped correctly: (16 × 7/8) pops to 14. The popping was fine. The problem was inside the circle. The 7/8 was only an estimate, and it was a little too big. A pop can only be as good as the numbers inside the circle.

STEP 3 The trap

Two friends, Jo and Sam, each mark 16 pegs. Both of them place every mark 1/32 of an inch too far, the same small slip each time.

Jo measures every mark from the zero end of the board. Sam measures each mark from the one before it.

At the 16th mark, how far off is each friend?

Jo is off by 1/32 of an inch, because each of her marks has only its own slip. Sam is off by 16 times 1/32, which is a full half inch, because every slip got carried along to the next mark.

Same slip, very different boards. The only difference is where each friend measured from.

WHAT DID WE KEEP?

Write the answers in your notebook.

- When Mara fixed the marks, **what stayed the same?** (Every mark had the same kind of mistake, just more of it each time. One number fixes them all.)
- **What changed?** (How big the mistake was at each mark.)

WHAT MAKES THIS CORRECTION TRUSTWORTHY?

There are three kinds of fixes in this chapter. Which is which?

1. Moving each mark back by n/8 because the spacing is exactly 7/8.
2. Using a guess of 7/8, then checking the 16th mark against the sample board.
3. Measuring from the zero end instead of from the previous mark.

(Answers: 1 is an exact fix, because the numbers are known. 2 is an estimate that needs a check. 3 is a habit that keeps small errors from piling up.)

PRACTICE

1. Mara's marks were supposed to be 3/4 inch apart but she placed them 1 inch apart. How far back must mark 6 move? Where does it belong?

2. This time the marks were supposed to be 1 inch apart, but she placed them 3/4 inch apart. They are too short. At which mark is the board a full foot short?

3. Mara guesses that her spacing is 7/8 inch. Her guess says mark 20 belongs at 17 1/2 inches. The sample board has its 20th peg at 17 1/4 inches. Where should mark 40 be? (Hint: if the mistake grows steadily, what happens when the number of marks doubles?)

4. Marks that should be 7/8 inch apart were placed 1 inch apart. Write the correction for mark 12 as a circle, redraw it, and pop it.

BRIDGE *Bridge to The Work That Vanishes:* Mara did not have to remeasure every mark from scratch. She named one small mistake, counted how many times it happened, and fixed everything at once. The multiplication tricks in that book work the same way: take the easy answer, name the leftover piece, and add it back or take it away.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **899**. Leave the others blank. What appears?

890 + 12	881 + 18	929 − 30	880 + 12	884 + 6	863 + 36	924 − 25	871 + 23
937 − 38	885 + 14	893 + 6	922 − 23	884 + 15	902 − 3	916 − 17	866 + 33
889 + 10	881 + 18	877 + 22	868 + 31	916 − 17	928 − 29	906 − 7	881 + 18
920 − 21	901 − 2	887 + 12	911 − 12	914 − 15	883 + 16	865 + 34	915 − 16
884 + 6	895 + 4	884 + 15	871 + 28	902 − 3	891 + 8	910 − 11	93 × 93 − 88 × 88
881 + 11	872 + 24	914 − 15	866 + 33	911 − 12	913 − 14	11 × 82	884 + 3
871 + 25	887 + 6	56 × 16	904 − 5	31 × 29	880 + 25	898 + 2	880 + 13
113 × 113 − 109 × 109	874 + 28	889 + 16	865 + 29	2 × 451	883 + 19	868 + 27	875 + 27



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 2

Close Enough, and How Close

STORY

Theo sold apples at a market stall. One morning a customer asked, “How many apples do you have in your 49 crates, if each crate holds 21?”
Theo had no calculator. But he noticed that 49 is nearly 50, and 21 is nearly 20, and 50 × 20 is easy.

STEP 1 An easy answer that is a little wrong

Theo drew one circle and popped it:

TRACE

```
( 50 × 20 )  →  1,000
```

“About a thousand,” he said.

That is a useful wrong answer. It is wrong, but it is wrong by an amount Theo can work out. To find how wrong, he moved one number at a time and wrote down what each move did.

Move 1: change 50 back to 49. That is one crate fewer, and a crate holds 20 in his estimate, so it takes away 20.

Move 2: change 20 back to 21. Now each of the 49 crates holds one more apple, so it adds 49.

Here is the notebook. Every redraw keeps the value, and every pop is easy:

TRACE

```
( 49 × 21 )
redraw  →  ( ( 49 × 20 ) + ( 49 × 1 ) )
redraw  →  ( ( ( 50 × 20 ) − ( 1 × 20 ) ) + ( 49 × 1 ) )
pop ( 50 × 20 )  →  1,000
pop ( 1 × 20 )   →  20
pop ( 49 × 1 )   →  49
( ( 1,000 − 20 ) + 49 )
pop ( 1,000 − 20 )  →  980
( 980 + 49 )
pop  →  1,029
```

So Theo’s estimate had two named errors: **minus 20** and **plus 49**. Together they come to plus 29, and 49 × 21 is **1,029**.

Trust check. There are three different kinds of answer in this chapter.

STEP 2 Which error matters?

The two errors pull in opposite directions, so you cannot tell whether the estimate is too low or too high until you compare their sizes. For 49 × 21, the plus 49 is bigger than the minus 20, so the estimate is too low.

Try 51 × 19. Starting again from (50 × 20), which is 1,000:

- 51 is one more crate of 20, so **plus 20**.
- 19 is one fewer apple in each of 51 crates, so **minus 51**.

1,000 + 20 − 51 = **969**. This time the estimate is too high, by 31.

STEP 3 A range without the exact answer

Sometimes you do not need the exact number. You only need to know a range that is certainly right.

49 × 21 is at least 49 × 20, because 21 is more than 20. And it is at most 50 × 21, because 49 is less than 50. Two easy circles give the ends:

TRACE

```
( 49 × 20 )  →  980
( 50 × 21 )  →  1,050
```

So the circle (49 × 21) pops to something between 980 and 1,050. That is a **bounded estimate**. It says less than the exact answer, but it is always right.

A bound is only useful if it is tight enough for the job. Suppose a truck can carry 1,000 apples. The bounds 980 to 1,050 cannot decide whether the truck is enough, because 1,000 sits inside the range. Only the exact answer, 1,029, settles it: the truck is not big enough.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the estimate and its named errors keep?** (Everything. Add the errors back and you get the exact answer.)
- **What did the bound keep, and what did it lose?** (It kept a range that is certainly right. It lost the exact value.)

WHAT MAKES THIS TRUSTWORTHY?

There are three different kinds of answer in this chapter.

1. **An exact fix.** Every correction is counted, so the answer is exact. This is a reason, because each redraw keeps the value.
2. **A bound.** The ends come from simple facts, such as 49 being less than 50. This is a reason too, but it only tells you a range.
3. **A plain estimate with no errors named.** "About a thousand" is only a guess until someone says how far off it could be.

A good habit is to check with a different route. Here, $50 \times 21 - 21$ is $1{,}050 - 21$, which is 1,029. The routes agree.

PRACTICE

1. Start from (50×20). Name the two corrections for **51×19**, then find the exact answer.

2. Start from (100×10). Name the two corrections for **99×11**, then find the exact answer.

3. Find a lower bound and an upper bound for **38×52**, using easy circles.

4. A truck carries 1,100 apples. Do the bounds 980 to 1,050 decide whether it is big enough for 49 crates of 21? What if the truck carries only 1,000?

5. Redraw (49×20) as ((50×20) − (1×20)) and pop every circle.

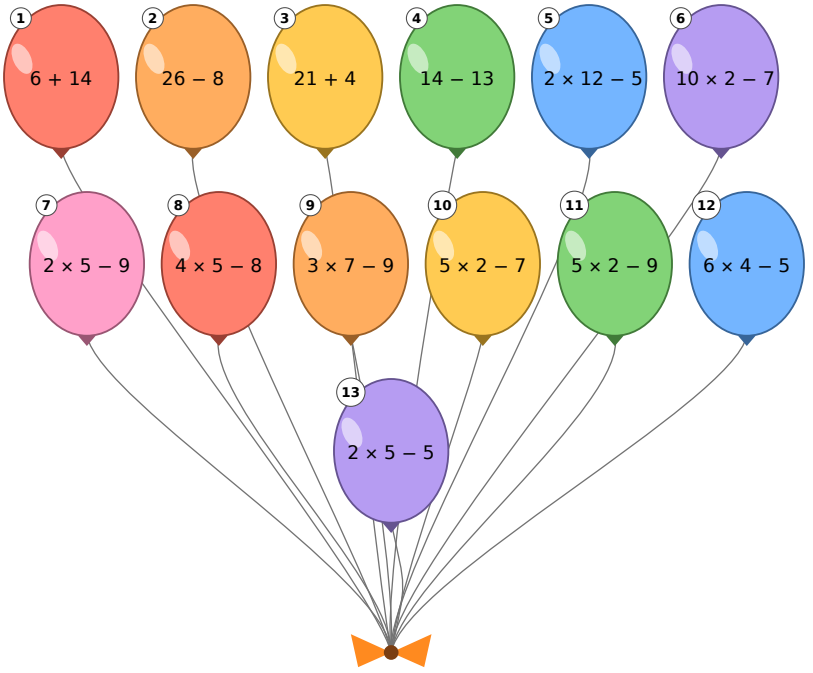
BRIDGE *Bridge to the rest of the series:* In chapter 1, Mara's mistake grew steadily with the mark number. Here the estimate has two separate errors, one small and one large, and naming both is what turns a guess into an answer.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

												!
1	2	3	4	5	6	7	8	9	10	11	12	13

CHAPTER 3

Two Cheap Checks

STORY

Theo's friend Nia helped at the stall. She worked out 23 crates of 47 apples and wrote **1,071** on a card. Theo's cousin Ola worked out the same thing and wrote **1,801**. A third helper wrote **1,081**.
Three answers, and only one of them can be right. Theo did not want to redo all the work three times. He wanted a quick way to throw out bad answers.

STEP 1 The size check

First Theo made an easy estimate with round numbers. He set a floor and a ceiling, as in chapter 2:

TRACE

```
( 20 × 40 )  →  800     (23 is more than 20, and 47 is more than 40, so the real answer is bigger)
( 25 × 50 )  →  1,250   (23 is less than 25, and 47 is less than 50, so the real answer is smaller)
```

So 23 × 47 must be somewhere between 800 and 1,250.

- Nia's 1,071 fits in the range.
- Ola's 1,801 does not. It is too big, so Ola's answer is **wrong**.
- The third helper's 1,081 fits.

The size check threw out one answer in a few seconds.

STEP 2 The last-digit check

Next Theo looked only at the last digit. In 23 × 47, the last digits are 3 and 7, and 3 × 7 = 21. The last digit of the real answer has to be the last digit of 21, which is **1**.

Why? When you multiply two numbers, the tens and hundreds parts of each number only change the tens and hundreds of the answer. They cannot change the ones digit.

- Nia's 1,071 ends in 1. It passes.
- Ola's 1,801 ends in 1. It passes, though it already failed the size check.
- The third helper's 1,081 ends in 1. It passes.

Trust check. The trust comes from the reason for each check.

Now Theo has a puzzle. Nia's 1,071 and the third helper's 1,081 both passed both checks. Only one of them can be right.

STEP 3 What a check can and cannot do

Here is the important thing about these checks.

- **When a check fails, you know the answer is wrong.** Ola's 1,801 failed the size check, so it is definitely wrong.
- **When a check passes, you do not know the answer is right.** Nia's 1,071 passed both checks, and it is wrong.

A check is like a net with holes. It catches big fish, such as the answer that is far too large or one with the wrong last digit. A small slip, like a 7 in the tens place instead of an 8, slips through the holes.

To settle it, Theo used the third kind of check: another route. This is the corner trick from The Work That Vanishes.

TRACE

```
( 23 × 47 )
redraw  →  ( ( 35 − 12 ) × ( 35 + 12 ) )
redraw  →  ( ( 35 × 35 ) − ( 12 × 12 ) )
pop  →  ( 1,225 − 144 )
pop  →  1,081
```

The answer is **1,081**. Nia's 1,071 was ten short, a slip in the tens place that the two cheap checks could not see.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does a failed check keep?** (Certainty: the answer is wrong.)
- **What does a passed check keep?** (Only a suspicion that the answer might be right.)

WHAT MAKES THIS TRUSTWORTHY?

The trust comes from the reason for each check. The size check works because a number between two easy circles must lie between their values. The last-digit check works because only the ones digits can reach the ones place. A check that you cannot give a reason for is not a check. It is a hunch. And a second route, with its own reason, is the only way to turn a "might be right" into "is right".

PRACTICE

1. Use the size check to say whether 38 × 52 could be **2,300**. (Use 38 × 50 and 40 × 52 as the floor and ceiling, as in chapter 2.)
2. What must the last digit of 38 × 52 be? Could the answer be **1,975**?
3. A card says 38 × 52 = **1,986**. Does it pass both checks? Is it right? (Hint: the real answer is 1,976.)
4. Find the last digit of 64 × 75. What does that tell you about **4,805**?
5. Dani says: "My answer passed both checks, so it is correct." What would you say?

BRIDGE *Bridge to the rest of the series:* In chapter 2, a bound gave a range that was certainly right. Here that range is used as a net to catch wrong answers. The second route in Step 3 is the missing-corner trick from The Work That Vanishes, used as a check. In Two Boxes, One Answer, Box P looked right too, and a missing check was the problem.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  9
6 pop ( 1,600 − 9 )  →  1,591
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × 1 ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × 1 )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × ( 6 + 1 ) ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × ( 6 + 1 ) )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 4

Small Compared With What

STORY

Theo made two estimates on the same day, and both were wrong by exactly 8.
Estimate 1. Four crates of 98 apples each. Theo said "about 400." The exact answer is 392. He was off by 8.
Estimate 2. A small bag of apples that really held 12. Theo said "about 20." He was off by 8.
The errors were the same size. But Theo felt that the second one was much worse. Was he right?

STEP 1 Compare the error with the answer

An error of 8 means very little when the answer is 392. It means a great deal when the answer is only 12. So Theo asked, "How big is the error **compared with the answer**?"

TRACE

```
8 out of 392   →   about 2 out of 100
8 out of 12    →   about 2 out of 3
```

Check the first one. Two out of every hundred of 392 is about 7.84, which is close to 8. ✓

The second one is more than half. Theo's estimate of 20 was not even close: it was nearly twice the real 12.

So the same error can be tiny or huge. The size of an error only means something when you say **what it is small compared with**.

STEP 2 Where does the answer come from?

Theo's error of 8 came from rounding 98 up to 100 in each crate. Here is the named error, as in chapter 2:

Trust check. The error is named by the redraw, so it is exact: 4 × 2.

TRACE

```
( 4 × 98 )
redraw  →  ( ( 4 × 100 ) − ( 4 × 2 ) )
pop  →  ( 400 − 8 )
pop  →  392
```

The error is **4 × 2 = 8**. For 4 crates it is small. If Theo had 400 crates, the same rounding would give an error of 400 × 2 =

800, but the answer would be 400 × 98 = 39,200. The error grew, but the *share* stayed the same:

TRACE

```
8 out of 392       →  about 2 out of 100
800 out of 39,200  →  about 2 out of 100
```

Rounding 98 to 100 is always an error of about 2 in 100, however many crates there are.

STEP 3 What is good enough?

Whether an error is acceptable depends on the job.

- Ordering apples for a stall? An error of 2 in 100 is fine.
- Counting the exact change owed to a customer? Even an error of 1 is not.
- Deciding whether a truck holds enough? It depends on how close the answer is to the truck's limit, as in chapter 2.

There is no number that is "small enough" for every job. What matters is the job, and a named error lets you decide.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same when the number of crates grew?** (The share of the error, about 2 in 100.)
- **What grew?** (The error itself, and the answer.)

WHAT MAKES THIS TRUSTWORTHY?

The error is named by the redraw, so it is exact: 4 × 2. The comparison with the answer is a separate step, and it is an estimate itself. "About 2 in 100" is a rough measurement, and it needs its own care: always say what you compared with.

PRACTICE

1. Theo says "about 1,000" for 49 × 21, and the real answer is 1,029. The error is 29. About how many in 100 is that? (Is it closer to 3 or to 30?)

2. An estimate of 600 is off by 31 from the real answer of 569. Is the share closer to 5 in 100 or 50 in 100?

3. Two estimates are both off by 5. One is for an answer of 500 and one for an answer of 10. Which is the worse estimate and why?

4. Rounding 99 to 100 gives an error of 1 in each. For 7 crates of 99, what is the exact answer, the error, and about what share?

5. Dani says: “An error of 8 is small.” What question should you ask Dani?

BRIDGE *Bridge to the rest of the series:* In chapter 3, the checks decided whether an answer could be right. Here the named error decides how good a wrong answer is. In The Boundary of a Step, an estimate is permitted for a rough idea and refused for an exact bill.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,400**. Leave the others blank. What appears?

18 × 134	2374 + 26	4 × 600	2378 + 24	2386 + 13	2391 + 9	10 × 240	4 × 603
2365 + 35	62 × 62 − 38 × 38	70 × 70 − 50 × 50	106 × 106 − 94 × 94	2370 + 30	2362 + 38	2367 + 33	2364 + 36
160 × 15	2434 − 34	49 × 49 − 1 × 1	2391 + 9	2396 + 4	2433 − 33	150 × 16	2374 + 26
2425 − 25	2381 + 19	24 × 100	2390 + 10	2430 − 30	2370 + 30	2431 − 31	2387 + 13
2386 + 8	125 × 125 − 115 × 115	2410 − 10	2408 − 8	2419 − 19	2380 + 20	2382 + 18	2375 + 29
2395 + 15	2361 + 27	2361 + 39	120 × 20	2427 − 27	55 × 55 − 25 × 25	2371 + 28	2374 + 28
2379 + 9	2394 + 8	2387 + 14	2392 + 8	160 × 15	2384 + 20	2399 + 5	2393 + 2
2377 + 13	2369 + 23	2361 + 28	5 × 479	2404 + 4	2389 + 17	2393 + 9	2390 + 12



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 5

Errors That Cancel

STORY

Theo wanted the total of four crate counts: 47, 38, 52 and 41. He rounded each to the nearest ten, because tens are easy to add.

TRACE

```
( 50 + 40 + 50 + 40 )  →  180
```

The exact total is 178. So Theo's estimate is off by 2. That is surprisingly close for four roundings, and he wondered why.

STEP 1 Track every error

Theo wrote the error of each rounding. In this book, **error = estimate minus exact**, so a rounding up has a plus and a rounding down has a minus.

Number	Rounded	Error
47	50	+3
38	40	+2
52	50	−2
41	40	−1

Add the errors: +3 + 2 − 2 − 1 = **+2**. The estimate is 2 too high.

The errors pulled in both directions. The plus errors and the minus errors partly cancelled each other, so the total error is small.

In circles:

TRACE

```
( ( 47 + 38 ) + ( 52 + 41 ) )
redraw  →  ( ( 50 + 40 + 50 + 40 ) − ( ( 3 + 2 ) − ( 2 + 1 ) ) )
pop  →  ( 180 − 2 )
pop  →  178
```

Trust check. The error of each rounding is exact, so the total error is exact too: it is the sum of the individual errors.

STEP 2 When the errors do not cancel

Now a different list: 44, 43, 42 and 41. Round each to the nearest ten.

TRACE

```
( 40 + 40 + 40 + 40 )  →  160
```

The exact total is 170. Every number was rounded **down**, so every error is a minus: −4, −3, −2, −1. They all pull the same way, and they add up to −10.

Rounding the first list gave an error of 2. Rounding this list gave an error of 10. The same method, but the first time the errors cancelled and this time they stacked.

STEP 3 What to watch for

Errors cancel when some numbers round up and some round down. They stack when most numbers sit on the same side of a round number.

A careful estimator asks two questions.

- Do the numbers round in **both directions**? Then the errors will probably partly cancel.
- Do they all round **the same way**? Then expect a larger error, and be ready to correct for it.

Notice the word “probably”. Cancelling is not a promise. A list can round up and down and still leave a big total error if the ups are much bigger than the downs.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the estimate keep when errors cancelled?** (Nearly the whole total.)
- **What did it lose when errors stacked?** (A noticeable part, 10 out of 170.)

WHAT MAKES THIS TRUSTWORTHY?

The error of each rounding is exact, so the total error is exact too: it is the sum of the individual errors. That is a reason. The advice “errors tend to cancel” is only a tendency, so check by adding the errors whenever it matters.

PRACTICE

1. Round 63, 58, 71 and 49 to the nearest ten and add. Then find the exact total and the error.
2. Write the four errors for problem 1. Do they add to the total error?
3. Round 34, 33, 32 and 31 to the nearest ten. Do the errors cancel or stack? What is the total error?
4. Make a list of three numbers whose rounding errors cancel exactly to 0.
5. Theo says "rounding always gives a small error." What would you tell him?

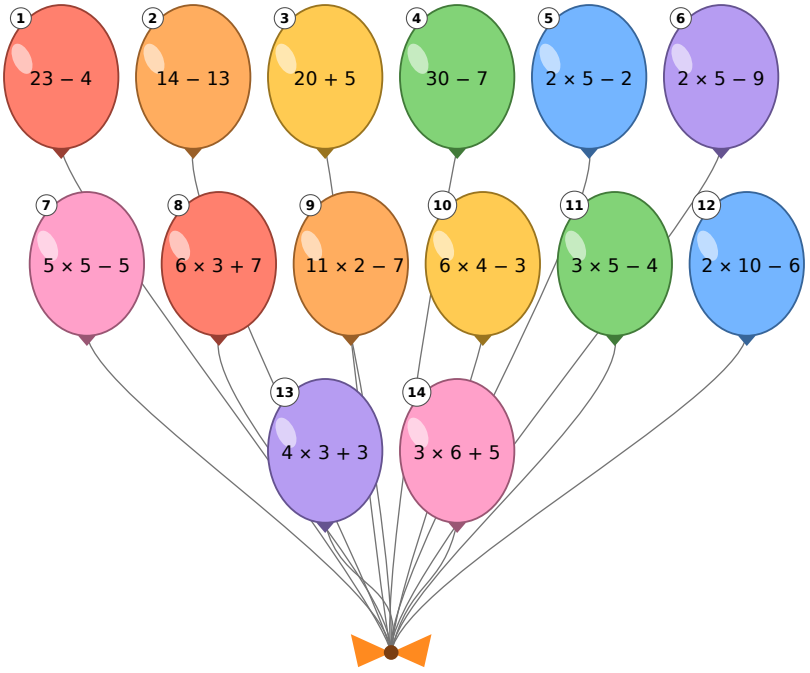
BRIDGE *Bridge to the rest of the series:* In chapter 4, an error was small or large compared with the answer. Here each rounding has its own error, and the total error is their sum. In The Work That Vanishes, the strips and the shared corner are errors that have to be added back in the right direction.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 6

Guess the Average, Then Correct the Guess

STORY

Theo sold apples for four days: 90, 85, 100 and 81 on the four days. What was the average number sold per day? The usual way is to add them all and divide by 4. Theo tried something different. He made a guess, and then corrected it.

STEP 1 Guess, then measure how far off

Theo guessed that the average was **90**. Then he checked each day against the guess.

Day	Sold	Compared with 90
1	90	0
2	85	5 below
3	100	10 above
4	81	9 below

The days above the guess add up to 10. The days below add up to 5 + 9 = 14. So overall, the four days are 14 − 10 = **4 below** the guess.

If four days are 4 below in total, then each day is 4 ÷ 4 = **1 below** on average. So the average is 90 − 1 = **89**.

In circles:

TRACE

```
( ( 90 + 85 + 100 + 81 ) ÷ 4 )
```

becomes

TRACE

```
( 90 − ( ( 14 − 10 ) ÷ 4 ) )
pop ( 14 − 10 )  →  4
pop ( 4 ÷ 4 )    →  1
pop ( 90 − 1 )   →  89
```

Check by the usual way: 90 + 85 + 100 + 81 = 356, and 356 ÷ 4 = 89. ✓

STEP 2 A good guess makes the correction small

Suppose the four days had been 78, 82, 85 and 75, and Theo guessed 80.

- 78 is 2 below.
- 82 is 2 above.
- 85 is 5 above.
- 75 is 5 below.

Above: 2 + 5 = 7. Below: 2 + 5 = 7. They cancel exactly, so the four days are 0 away from the guess. The average is **80**. Check: 78 + 82 + 85 + 75 = 320, and 320 ÷ 4 = 80. ✓

Trust check. The reason is that every day's difference from the guess is counted, above and below, and the total difference is shared among the days.

STEP 3 A bad guess still works

What if Theo had guessed 100 for the first list? Every day is at or below 100, by 10, 15, 0 and 19. That is 44 below in total, and 44 ÷ 4 = 11 below on average.

TRACE

```
( 100 − 11 )  →  89
```

Same answer, 89. A bad guess just makes a bigger correction.

This is Split, Balance, Correct again. The guess is the balance, the differences are the split, and the correction puts back what the guess got wrong.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same whatever the guess?** (The average, 89.)
- **What changed?** (The size of the correction.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that every day's difference from the guess is counted, above and below, and the total difference is shared among the days. Nothing is left out. The guess can be anything, because the correction repairs whatever the guess missed.
The check is the plain way: add everything and divide.

PRACTICE

1. Find the average of 72, 68, 75 and 65 with a guess of 70.
2. Find the average of 34, 41, 38 and 47 with a guess of 40.
3. Find the average of 20, 30, 40 and 22 with a guess of 30.
4. Find the average of 12, 15, 18 and 15. What guess is easiest?
5. Dani guessed 50 for the first list (90, 85, 100, 81). What should the correction be, and what average does she get?

BRIDGE *Bridge to the rest of the series:* In chapter 5, errors from roundings could cancel or stack. Here the guess is deliberately off, and the differences above and below are the errors that get put back. In The Work That Vanishes, 50×50 as a guess for 48×52 and the corner as the correction are the same shape.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  4
6 pop ( 1,600 − 4 )  →  1,596
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × ( 4 + 1 ) ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × ( 4 + 1 ) )  →  20
6 pop ( 2,500 − 20 )  →  2,480
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 7

Counting Digits

STORY

Theo typed 23 × 47 into a calculator and the screen showed **10.81**. Then he typed it again and got **10,810**. He pressed the buttons one more time and got **1,081**. Three answers, three different sizes. Two of them have the digits right and the decimal point or a zero wrong. Which one is correct, and how could he tell at a glance?

STEP 1 How many digits should the answer have?

Theo knows that the smallest two-digit number is 10 and the largest is 99.

TRACE

```
( 10 × 10 )  →  100       (3 digits)
( 99 × 99 )  →  9,801     (4 digits)
```

So any two-digit number times a two-digit number lies between 100 and 9,801. That means the answer has **3 or 4 digits**, never 2 and never 5.

- 10.81 is not even a whole number. It cannot be right.
- 10,810 has 5 digits. It is too big. It cannot be right.
- 1,081 has 4 digits. It is possible.

The digit count threw out two of the three answers without any multiplying.

STEP 2 Narrow it with the floor and ceiling

Counting digits is a rough net. To narrow further, Theo uses the floor and ceiling from chapter 3.

Trust check. The digit count comes from the smallest and largest possible products, and these are exact.

TRACE

```
( 20 × 40 )  →  800
( 25 × 50 )  →  1,250
```

So 23 × 47 is between 800 and 1,250. The answer 1,081 sits comfortably inside.

The digit count alone could not decide between 3 and 4 digits here, since 800 has 3 digits and 1,250 has 4. The floor and ceiling made it clear.

STEP 3 Digits for other sizes

The same idea works for any sizes.

Multiplying	Smallest product	Largest product	Digits in the answer
2-digit × 2-digit	100	9,801	3 or 4
3-digit × 2-digit	1,000	98,901	4 or 5
2-digit × 1-digit	10	891	2 or 3

For the second row, the smallest is 100 × 10 and the largest is 999 × 99. For the third, the smallest is 10 × 1 and the largest is 99 × 9.

In the last row, the smallest product is 10 × 1 = 10, which has 2 digits, so the answer has 2 or 3 digits.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does the digit count keep?** (A rough size: how many places the answer must have.)
- **What does it lose?** (The exact value, and even the exact size when the range crosses a place.)

WHAT MAKES THIS TRUSTWORTHY?

The digit count comes from the smallest and largest possible products, and these are exact. It is a strong test for slips with decimal points and extra zeros, which are among the most common slips. It is a weak test for a slip in a middle digit, so, as always, a passed check is not a proof.

PRACTICE

1. A calculator shows 2,496, 249.6 and 24,960 for 48 × 52. Which could be right? Why?
2. How many digits can 12 × 13 have? Which of 15.6, 156 and 1,560 could be right?

3. How many digits can 305 × 12 have? (Use 300 × 10 and 310 × 12 or similar.)

4. A shop sells 99 packs of 99 sweets each. Could the total have 3 digits, such as 981? Is 9,801 sensible?

5. Dani says “I counted the digits and they match, so my answer is right.” What would you say?

BRIDGE *Bridge to the rest of the series:* In chapter 3, the floor and ceiling were a net for wrong answers. A digit count is an even rougher net, and it catches the slips that nets with finer holes can miss. In The Choice Before the Calculation, choosing the unit well makes decimal-point slips less likely.
Next in this book: Chapter 8: The Estimate That Owes Four.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **1,225**. Leave the others blank. What appears?

1205 + 29	1264 − 39	1259 − 34	72 × 17	38 × 38 − 15 × 15	17 × 53 + 18 × 18	1240 − 15	1216 + 21
27 × 43 + 8 × 8	1222 + 3	1200 + 25	1240 − 15	1254 − 29	26 × 44 + 9 × 9	1199 + 26	1215 + 10
1220 + 5	1261 − 36	1188 + 37	21 × 49 + 14 × 14	17 × 53 + 18 × 18	9 × 61 + 26 × 26	1218 + 7	4 × 66 + 31 × 31
1223 + 2	1207 + 18	1206 + 19	9 × 61 + 26 × 26	1260 − 35	1207 + 18	1202 + 23	25 × 45 + 10 × 10
1219 + 4	1200 + 25	1214 + 11	1233 − 8	1198 + 27	1193 + 32	1203 + 22	1193 + 29
1217 + 13	1202 + 27	1202 + 23	1244 − 19	1217 + 8	1195 + 30	1205 + 25	1207 + 12
1213 + 23	1214 + 14	1205 + 17	1193 + 32	1263 − 38	1208 + 27	1212 + 23	1207 + 8
1208 + 15	88 × 14	1192 + 22	1212 + 15	4 × 307	39 × 39 − 17 × 17	1208 + 15	1201 + 21



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

A USEFUL WRONG ANSWER

CHAPTER 8

The Estimate That Owes Four

STORY

Ravi was handed a card from the case file: **48 × 52**.
"About 2,500," he said at once.
"How wrong is that?" asked his teacher.
Ravi had no idea. He had an estimate, but he did not have a number for how far off it was. In this chapter he finds one, and he finds it without doing the long multiplication.

STEP 1 Say what the estimate changed

Ravi's estimate came from changing the problem. He moved 48 up by 2 to make 50, and moved 52 down by 2 to make 50.

TRACE

```
( 48 × 52 )
estimate ( 50 × 50 )  →  2,500
```

An estimate is a loan. He borrowed an easy problem, and the difference between the easy problem and the real one is what he **owes**. From The Work That Vanishes, Ravi knows what that difference is when two numbers are the same distance from the middle: a small square corner, 2 × 2.

So the estimate owes 4:

TRACE

```
( 48 × 52 )  =  ( ( 50 × 50 ) − ( 2 × 2 ) )
```

Ravi can now say the estimate is **too high by 4**, before he has the exact answer. Paying the debt gives it:

TRACE

```
pop ( 2 × 2 )  →  4
( 2,500 − 4 )
pop  →  2,496
```

The error is the estimate minus the exact answer: 2,500 − 2,496 = **4**. A positive error means the estimate was too high.

Trust check. The estimate is trustworthy because its error is a calculated number, not a hope.

STEP 2 Compare three estimates

Ravi tried two other easy estimates for the same card.

Estimate	Value	Error (estimate − exact)	Can you name it before finishing?
50 × 52	2,600	**104** (too high)	Yes: 48 went up 2, and 2 × 52 = 104
48 × 50	2,400	**−96** (too low)	Yes: 52 went down 2, and 2 × 48 = 96
50 × 50	2,500	**4** (too high)	Yes: the corner, 2 × 2

All three errors can be named. The 50 × 50 estimate is the best of the three because its error is both small and easy.

That is what makes an estimate useful. A wrong answer is useful when you can say how wrong it is, and which way.

STEP 3 Small compared with what?

Is an error of 4 small? In a number around 2,500, yes. It is less than 2 in every 1,000. If the answer were 12, an error of 4 would be huge. So Ravi always asks *small compared with what*.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the estimate 50 × 50 keep?** (The rough size, and the knowledge of exactly what it changed.)
- **What did it lose?** (The exact answer, until the correction of 4 was paid.)

WHAT MAKES THIS TRUSTWORTHY?

The estimate is trustworthy because its error is a calculated number, not a hope. Ravi can name the error from the way the numbers moved: 2 down and 2 up gives a corner of 2 × 2. He did not need the exact answer to do it. After paying the debt, he can check by a second route, such as 48 × 50 + 48 × 2 = 2,400 + 96, which is also 2,496.

PRACTICE

1. Estimate **49 × 51** with 50 × 50. What does the estimate owe? What is the exact answer?
2. Estimate **47 × 53** with 50 × 50. What does it owe? What is the exact answer?
3. Estimate **46 × 54** with 50 × 50. What does it owe? What is the exact answer?
4. Estimate **38 × 42** with 40 × 40. What does it owe, and what is the exact answer?
5. Ravi says, “2,500 is exactly right for 48 × 52 because it is a round number.” What do you say?

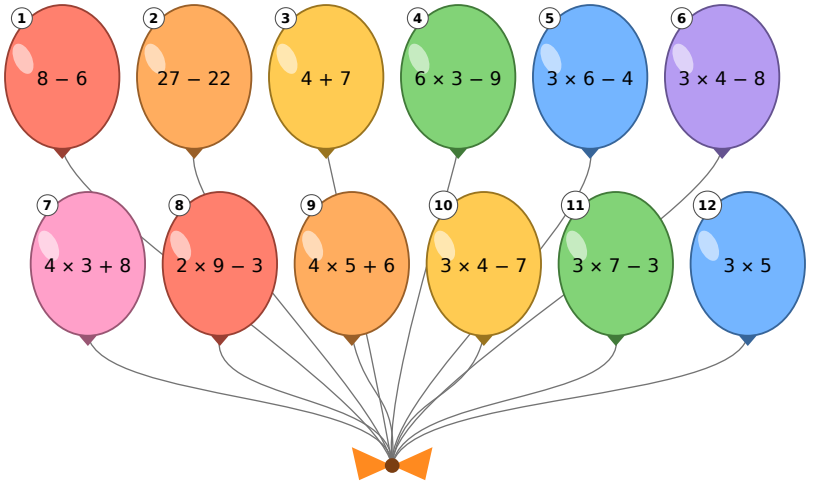
BRIDGE *Bridge to the rest of the series:* In The Work That Vanishes, the corner was found by cutting and sliding. Here the same corner is the debt on an estimate. In What Must Remain Unresolved, the debt is left unpaid for a while, and the notebook says so honestly.

PUZZLE PAGE

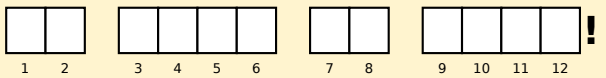
The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?



Answers (turn the book around)

Chapter 1

1. Move mark 6 back by 6/4, which is 1 1/2 inches. It belongs at 4 1/2 inches. Check: 6 times 3/4 is 4 1/2.
2. Each mark is short by 1/4 inch times its number. A foot is 12 inches, and 12 divided by 1/4 is 48. It is the 48th mark.
3. The guess was a quarter inch too long at mark 20. If the sample board's pegs are evenly spaced, it will be a half inch too long at mark 40. The guess says 35 inches, so the mark belongs at 34 1/2 inches.
4. The correction is (12 × (1 − 7/8)). Pop (1 − 7/8) to get 1/8, then (12 × 1/8) pops to **1 1/2**. Move mark 12 back by 1 1/2 inches.

Color by Answer (899). The shaded boxes make a heart.



Chapter 2

1. 51 is one more crate of 20, so **plus 20**. 19 is one less in each of 51 crates, so **minus 51**. 1,000 + 20 − 51 = **969**.
2. 99 is one fewer crate of 10, so **minus 10**. 11 is one more in each of 99 crates, so **plus 99**. 1,000 − 10 + 99 = **1,089**.
3. A lower bound is 38 × 50 = 1,900, because 52 is more than 50. An upper bound is 40 × 52 = 2,080, because 38 is less than 40. So the answer is between **1,900 and 2,080**. (The exact answer is 1,976.)
4. For a 1,100 truck, yes: the most it can be is 1,050, which fits. For a 1,000 truck, no: 1,000 is inside the range, so the bounds cannot decide.
5. (50 × 20) pops to 1,000 and (1 × 20) pops to 20, then (1,000 − 20) pops to **980**.

Balloon Message.

1.	6 + 14	= 20	T
2.	26 − 8	= 18	R
3.	21 + 4	= 25	Y
4.	14 − 13	= 1	A
5.	2 × 12 − 5	= 19	S
6.	10 × 2 − 7	= 13	M
7.	2 × 5 − 9	= 1	A
8.	4 × 5 − 8	= 12	L
9.	3 × 7 − 9	= 12	L
10.	5 × 2 − 7	= 3	C
11.	5 × 2 − 9	= 1	A
12.	6 × 4 − 5	= 19	S
13.	2 × 5 − 5	= 5	E

Message: **TRY A SMALL CASE!**

Answers (turn the book around)

Chapter 3

1. The floor is 38 × 50 = 1,900 and the ceiling is 40 × 52 = 2,080, so 2,300 is too big. It **cannot** be right.
2. 8 × 2 = 16, so the last digit must be **6**. The number 1,975 ends in 5, so it **cannot** be right.
3. It ends in 6, and it sits between 1,900 and 2,080, so it passes both checks. It is not right: the real answer is **1,976**.
4. 4 × 5 = 20, so the last digit is **0**. The number 4,805 ends in 5, so it is **wrong**. (The real answer is 4,800.)
5. Passing checks does not make an answer right. It only means these checks did not catch a mistake. Dani needs another route, such as a different way to split the multiplication.

Spot the Slip.
Notebook A: **perfect**. 37 × 43 = 1,591.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 399.
Notebook C: slip on line **3**. The corner is 6 × 6, not 6 × (6 + 1). The answer should be 4,864.

Chapter 4

1. 29 out of 1,029 is about 3 out of 100, so it is closer to **3**.
2. 31 out of 569 is about 5 out of 100, so it is closer to **5 in 100**.
3. The estimate for the answer of 10. Five out of 10 is half of the answer, while 5 out of 500 is 1 in 100.
4. ((7 × 100) − (7 × 1)) pops to 700 − 7 = **693**. The error is **7**, and 7 out of 693 is about **1 in 100**.
5. “Small compared with what?”

Color by Answer (2,400). The shaded boxes make a heart.



Answers (turn the book around)

Chapter 5

1. Rounded: 60 + 60 + 70 + 50 = **240**. Exact: 63 + 58 + 71 + 49 = **241**. The estimate is 1 too low.
2. Errors: 60 − 63 = −3, 60 − 58 = +2, 70 − 71 = −1, 50 − 49 = +1. They add to −3 + 2 − 1 + 1 = **−1**. Yes.
3. All round down, so they **stack**: −4, −3, −2, −1 add to **−10**. (Estimate 120, exact 130.)
4. For example 14, 16 and 20: 14 rounds to 10 (−4), 16 rounds to 20 (+4), 20 stays 20 (0). The errors add to 0. Many other lists work.
5. Rounding can give a small error or a big one. It depends on whether the errors cancel or stack, and on how big the answer is.

Balloon Message.

1.	23 − 4	= 19	S
2.	14 − 13	= 1	A
3.	20 + 5	= 25	Y
4.	30 − 7	= 23	W
5.	2 × 5 − 2	= 8	H
6.	2 × 5 − 9	= 1	A
7.	5 × 5 − 5	= 20	T
8.	6 × 3 + 7	= 25	Y
9.	11 × 2 − 7	= 15	O
10.	6 × 4 − 3	= 21	U
11.	3 × 5 − 4	= 11	K
12.	2 × 10 − 6	= 14	N
13.	4 × 3 + 3	= 15	O
14.	3 × 6 + 5	= 23	W

Message: **SAY WHAT YOU KNOW!**

Chapter 6

1. Differences: 2 above, 2 below, 5 above, 5 below. They cancel, so the average is **70**. (Sum 280, divided by 4.)
2. Differences: 6 below, 1 above, 2 below, 7 above. Above: 8. Below: 8. The average is **40**.
3. Differences: 10 below, 0, 10 above, 8 below. Net 8 below, shared among 4 days is 2 below. The average is 30 − 2 = **28**.
4. A guess of 15 works well. Differences: 3 below, 0, 3 above, 0. They cancel, so the average is **15**.
5. Every day is above 50 by 40, 35, 50 and 31, which is 156 in total. 156 ÷ 4 = 39. So 50 + 39 = **89**.

Spot the Slip.
Notebook A: **perfect**. 38 × 42 = 1,596.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 624.
Notebook C: slip on line **3**. The corner is 4 × 4, not 4 × (4 + 1). The answer should be 2,484.

Answers (turn the book around)

Chapter 7

1. Only **2,496** has 4 digits and is a whole number in the range 100 to 9,801. 249.6 is not whole and 24,960 has 5 digits.
2. 12 × 13 is between 100 and 9,801, so 3 or 4 digits. **156** is the only one of the three that fits (and 12 × 13 is exactly 156).
3. 305 × 12 is more than 300 × 10 = 3,000 and less than 310 × 12 = 3,720, so it has **4 digits**.
4. No. A 2-digit by 2-digit product has 3 digits only when it is below 1,000, and 99 × 99 is far above that, so 981 is too small. **9,801** is sensible, because 99 × 99 is 9,801.
5. Matching digits only catches slips in the size. A slip in a middle digit has the same number of digits. She needs another check.

Color by Answer (1,225). The shaded boxes make a heart.



Chapter 8

1. The numbers are 1 from the middle, so it owes 1 × 1 = **1**. The exact answer is 2,500 − 1 = **2,499**.
2. The numbers are 3 from the middle, so it owes 3 × 3 = **9**. The exact answer is 2,500 − 9 = **2,491**.
3. The numbers are 4 from the middle, so it owes 4 × 4 = **16**. The exact answer is 2,500 − 16 = **2,484**.
4. The middle is 40 and the distance is 2, so it owes 2 × 2 = **4**. The exact answer is 1,600 − 4 = **1,596**.
5. A round number is easy, but it is not the same as exact. 2,500 is too high by 4, and a good estimate says so.

Balloon Message.

1.	8 − 6	= 2	B
2.	27 − 22	= 5	E
3.	4 + 7	= 11	K
4.	6 × 3 − 9	= 9	I
5.	3 × 6 − 4	= 14	N
6.	3 × 4 − 8	= 4	D
7.	4 × 3 + 8	= 20	T
8.	2 × 9 − 3	= 15	O
9.	4 × 5 + 6	= 26	Z
10.	3 × 4 − 7	= 5	E
11.	3 × 7 − 3	= 18	R
12.	3 × 5	= 15	O

Message: **BE KIND TO ZERO!**